

UNIT 5 · TOPIC 5.5

AI and Machine Learning in Ticketing

Smarter prices, smarter service, smarter forecasts.



B.B.A. Aviation Management · Semester III · Unit 5: Technology & Trends
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TOPIC 5.5

Mission Video Briefing



Air Ticketing & Fare Construction · Unit 5: Technology & Trends · Topic 5.5

02

TOPIC 5.5

Mission Question



THINK ABOUT IT

Is the price you see set by a person — or by a machine?

**THINK**

30 seconds alone

**PAIR**

1 minute with a neighbour

**SHARE**

2-3 answers with the class



TOPIC 5.5

Mission Briefing

1

5.5.1

Role of AI in dynamic pricing and fare optimisation

2

5.5.2

Machine learning for personalised travel recommendations

3

5.5.3

AI chatbots and virtual assistants in customer service

4

5.5.4

Predictive analytics for demand forecasting and inventory



MISSION OBJECTIVES

- Explain AI in pricing
- Describe ML recommendations
- Evaluate AI chatbots
- Apply predictive analytics concepts

Pedagogy: Learning Through Problem-Solving

TOPIC 5.5.1

AI in Dynamic Pricing



TOPIC 5.5.1

Fare Optimisation Problems AI Solves

**Empty seats**

When to discount to fill seats without losing revenue.

**Competitor moves**

React to rival fare changes in minutes.

**Segments**

Different offers for leisure vs business demand.

**PROBLEM**

A flight is 60% full 10 days out; last year it reached 95%. Should the airline drop or hold prices? What data would an AI model use to decide?



TOPIC 5.5.2

Personalised Recommendations



Past Trips

Suggest routes and times you usually prefer.



Preferences

Aisle seat, veg meal pre-selected.



Destinations

“People like you also visited...”



Add-ons

Relevant upgrades or bags offered.

TOPIC 5.5.2

Recommendation Logic (Simplified)

How an ML model might score offers.

- Inputs: profile, history, context.
- Model scores each offer.
- Highest-scoring offers shown first.
- Feedback retrains the model.



RECOMMENDER (concept)

```
USER  student, DEL, budget  
CONTEXT  searching DEL-GOI Dec
```

OFFER	SCORE
Saver + 15kg	0.82
Flexi	0.41
Business	0.07

```
-> show Saver first
```



TOPIC 5.5.3

AI Chatbots & Virtual Assistants



Examples

Air India's AI.g, IndiGo's 6Eskai, KLM's messaging assistants.



24/7

Instant answers on web, app and WhatsApp.



Multilingual

Serve Hindi, Tamil, English and more.



Human Hand-off

Complex cases escalated to agents.



TOPIC 5.5.3

Chatbot vs Human Agent



Chatbot Best At

- Booking status, baggage FAQs
- Simple changes, web check-in
- Instant, unlimited scale

VS



Human Best At

- Complex itineraries
- Emotional situations
- Exceptions and judgement



TOPIC 5.5.4

Predictive Analytics in Action (Illustrative)

95%

forecast load factor for a festival flight

–15%

cheap seats released vs normal

+12%

expected revenue gain from better forecasting



Illustrative numbers — forecasting helps decide how many seats to sell at each price.

TOPIC 5.5.4

What Gets Predicted?

Prediction	Used for
Demand per flight	Seat allocation by fare class
No-shows	Overbooking levels
Cancellations	Inventory release
Price sensitivity	Discount decisions
Disruptions	Proactive rebooking

TOPIC 5.5

Innovation Lab: Be the Algorithm

**PEDAGOGY****Learning Through
Problem-Solving****TIME · 12 min****FORMAT · Groups of 4****1**

Groups get booking data for one flight over 30 days.

2

Forecast final demand and decide price changes each week.

3

Compare your revenue with the “actual” result card.

4

Discuss what data would improve your forecast.



TOPIC 5.5*Click to reveal each answer*

System Check Quiz

Q1**AI in pricing mainly helps to...**

- A) Print tickets
- B) Set optimal prices
- C) Clean aircraft
- D) Assign gates

ANSWER**B****Q2****Predicting no-shows supports...**

- A) Overbooking decisions
- B) Meal design
- C) Uniforms
- D) Parking

ANSWER**A****Q3****Chatbots are best for...**

- A) Emotional crises
- B) Simple FAQs and status checks
- C) Complex RTW fares
- D) Legal disputes

ANSWER**B****Q4****ML recommendations use...**

- A) Random choice
- B) Customer data and patterns
- C) Only price
- D) Weather only

ANSWER**B**

MISSION COMPLETE

Key Takeaways

- ✓ AI optimises prices using demand and competitor data
- ✓ ML personalises recommendations and add-ons
- ✓ Chatbots handle routine service; humans handle complexity
- ✓ Predictive analytics drives inventory and overbooking

EXIT TICKET Write 1 thing you learned today + 1 question you still have.

NEXT MISSION

**Topic 5.6 — Mobile
Ticketing Solutions**

Thank you · Questions?